Inter-VLAN Routing Using Router-on-a-Stick

Cisco Packet Tracer Networking Project

1. Project Overview

Objective: Configure three separate VLANs (HR, IT, Sales) and enable communication between them using Router-on-a-Stick.

Software: Cisco Packet Tracer

Devices:

- 1 Router
- 1 Switch
- 3 VLANs
- 7 PCs/Laptops

Technologies:

- VLAN
- Access Ports
- Trunking
- IEEE 802.1Q
- Router Subinterfaces
- Inter-VLAN Routing
- ICMP Ping Testing

2. Network Topology

The network consists of one router connected to one switch. The switch connects devices from three departments.

- VLAN 10 – HR
- VLAN 20 – IT
- VLAN 30 – Sales

The router provides communication between these VLANs.

3. IP Addressing Scheme

Department	VLAN	Network	Gateway
HR	10	10.10.10.0/24	10.10.10.1
IT	20	10.10.20.0/24	10.10.20.1
Sales	30	10.10.30.0/24	10.10.30.1

Subnet Mask: 255.255.255.0

End Device IP Addresses

Device	IP Address	Default Gateway
HR PC	10.10.10.2	10.10.10.1
HR Laptop	10.10.10.3	10.10.10.1
HR PC 2	10.10.10.4	10.10.10.1
IT Laptop	10.10.20.2	10.10.20.1
IT PC	10.10.20.3	10.10.20.1
Sales PC	10.10.30.2	10.10.30.1
Sales Laptop	10.10.30.3	10.10.30.1
Sales PC 2	10.10.30.4	10.10.30.1

4. VLAN Configuration on Switch

Created three VLANs:

enable

configure terminal

vlan 10

name HR

exit

vlan 20

name IT

exit

vlan 30

name Sales

exit

Assign Switch Ports

VLAN	Department	Ports
10	HR	Eth7/1, Eth8/1, Eth9/1
20	IT	Eth5/1, Eth6/1
30	Sales	Eth2/1, Eth3/1, Eth4/1

Verify VLANs

show vlan brief

Expected result: VLAN 10, VLAN 20 and VLAN 30 should appear with their assigned access ports.

5. Trunk Port Configuration

The router is physically connected to switch port Eth1/1.

Configured the switch port as a trunk:

enable

configure terminal

interface ethernet 1/1

switchport mode trunk

no shutdown

exit

end

Verify Trunk

show interfaces trunk

show interfaces status

Expected result:

- Port: Eth1/1
- Status: trunking
- Encapsulation: 802.1Q
- Active VLANs: 1, 10, 20, 30

6. Router-on-a-Stick Configuration

Router interface: GigabitEthernet0/0

Enabled the physical interface:

enable

configure terminal

interface gig0/0

no ip address

no shutdown

exit

VLAN 10 – HR Subinterface

interface gig0/0.10

encapsulation dot1Q 10

ip address 10.10.10.1 255.255.255.0

exit

VLAN 20 – IT Subinterface

interface gig0/0.20

encapsulation dot1Q 20

ip address 10.10.20.1 255.255.255.0

exit

VLAN 30 – Sales Subinterface

interface gig0/0.30

encapsulation dot1Q 30

ip address 10.10.30.1 255.255.255.0

Verify Router Interfaces

show ip interface brief

Expected result:

Interface	IP Address	Status
G0/0	Unassigned	Up/Up
G0/0.10	10.10.10.1	Up/Up
G0/0.20	10.10.20.1	Up/Up
G0/0.30	10.10.30.1	Up/Up

7. Configure End Device IP Addresses

For each PC or laptop:

1. Open the device.
2. Select Desktop.
3. Open IP Configuration.
4. Select Static.
5. Enter the assigned IP address.
6. Enter subnet mask 255.255.255.0.
7. Enter the correct default gateway.

Example for HR PC:

- IP Address: 10.10.10.2

- Subnet Mask: 255.255.255.0
- Default Gateway: 10.10.10.1

Repeat for all devices using the IP addressing table.

8. Connectivity Testing

Used the ping command to test communication.

Test 1: Same VLAN Communication

From HR PC:

ping 10.10.10.3

Expected result: Successful replies.

Test 2: HR to IT Communication

ping 10.10.20.2

Expected result: Successful replies.

Test 3: HR to Sales Communication

ping 10.10.30.2

Expected result: Successful replies.

Test 4: IT to Sales Communication

ping 10.10.30.2

Expected result: Successful replies.

Final Result: All tested pings were successful.

9. Verification Commands

The following commands were used to verify the network:

show vlan brief

show interfaces status

show interfaces trunk

show ip interface brief

show ip route

show running-config

10. Project Outcome

Successfully configured and tested a three-VLAN network using Router-on-a-Stick.

The project demonstrates:

- Department-wise network segmentation.
- VLAN creation and access port assignment.
- 802.1Q trunk configuration.
- Router subinterface configuration.
- Inter-VLAN routing.
- IP addressing and default gateway configuration.
- Network connectivity troubleshooting and verification.

11. Skills Learned

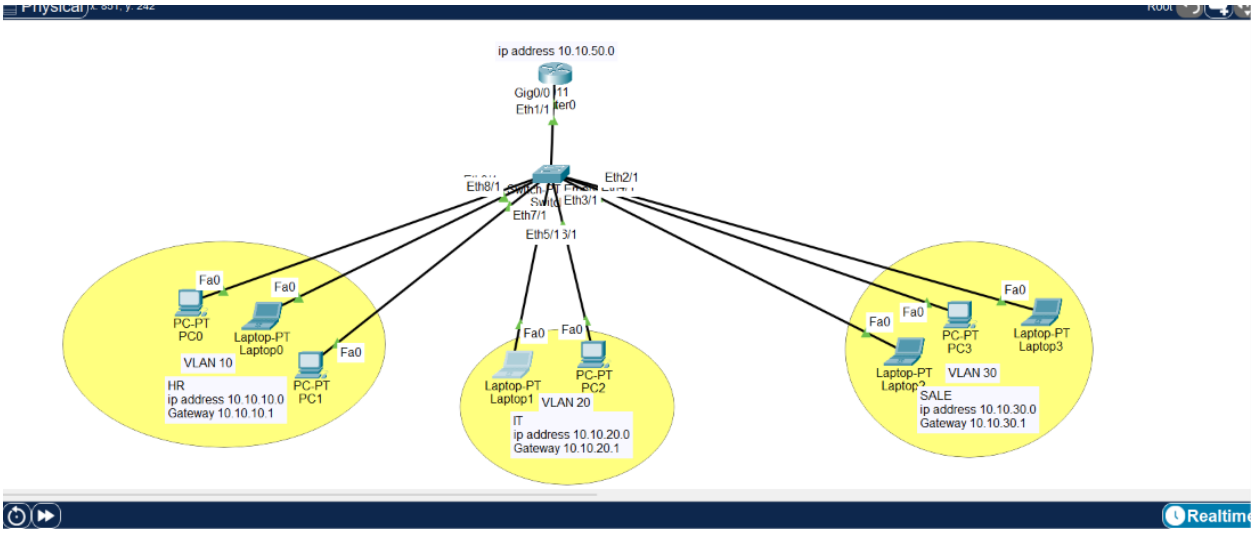
- Cisco IOS CLI
- VLAN Management
- Trunk Port Configuration
- Router-on-a-Stick
- Inter-VLAN Routing
- IP Subnetting
- Network Troubleshooting
- ICMP and Ping Testing
- Basic Network Documentation

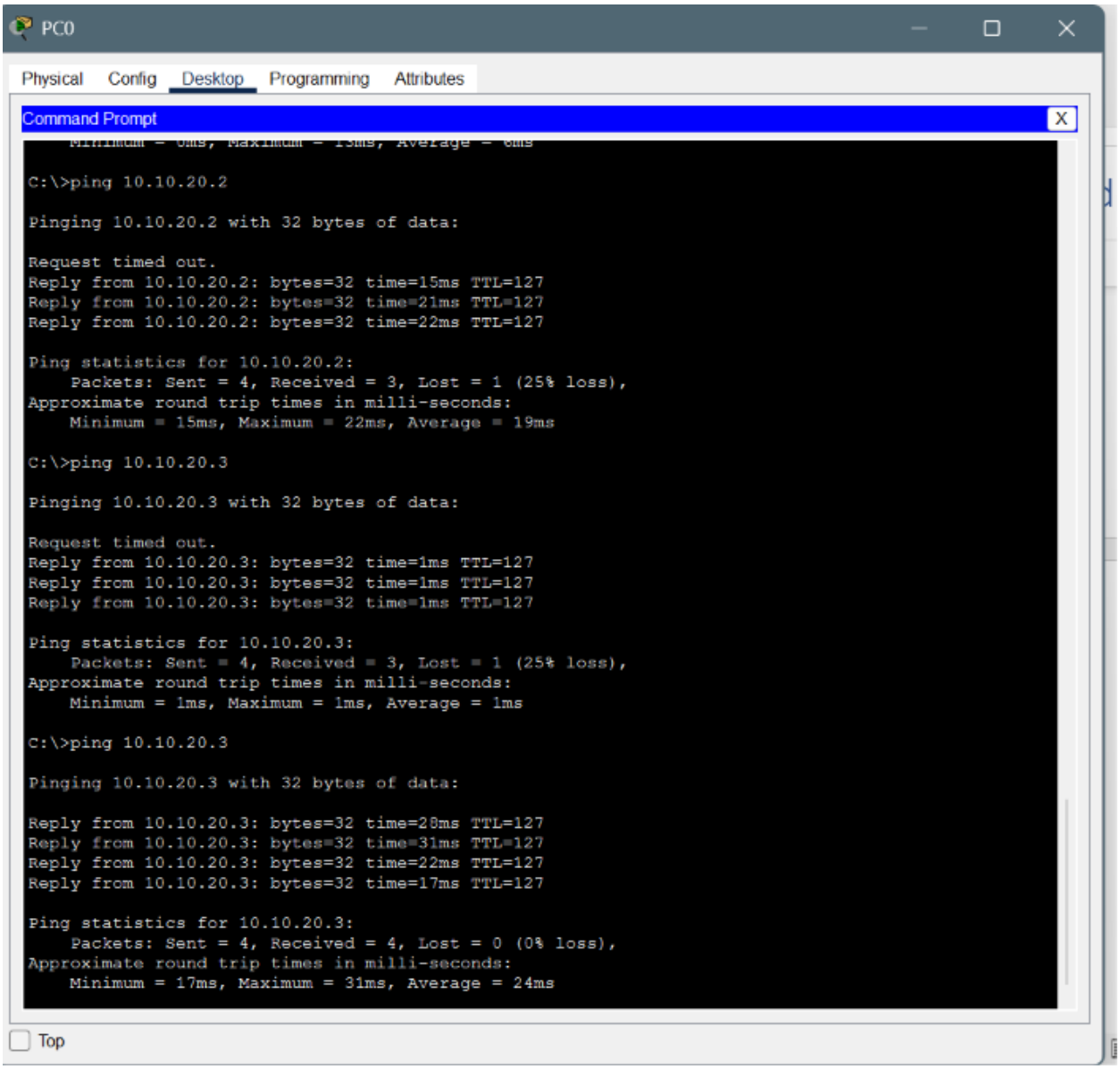
12. Conclusion

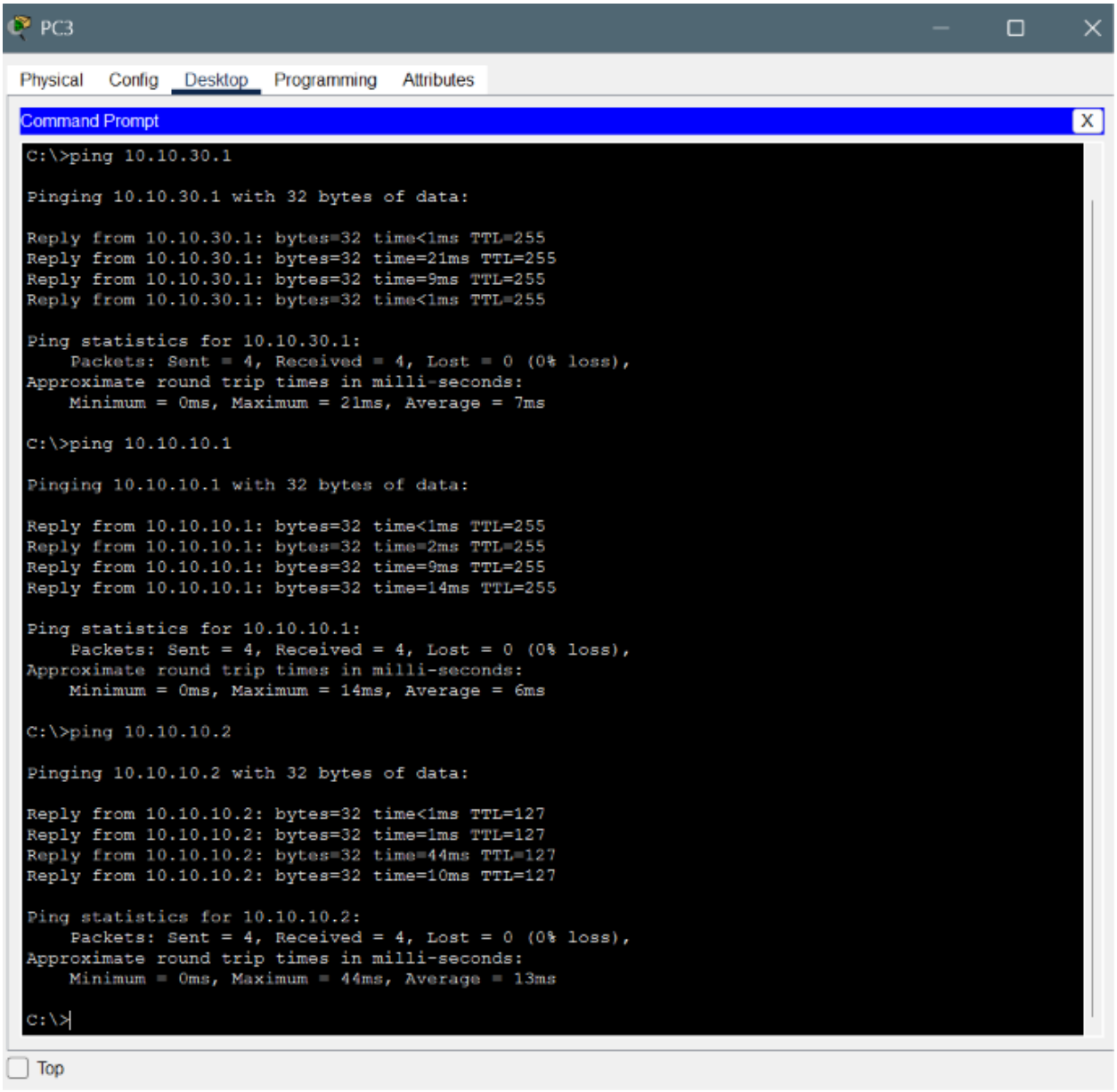
This project demonstrates how multiple VLANs can communicate through a single physical router interface using 802.1Q trunking and router subinterfaces.

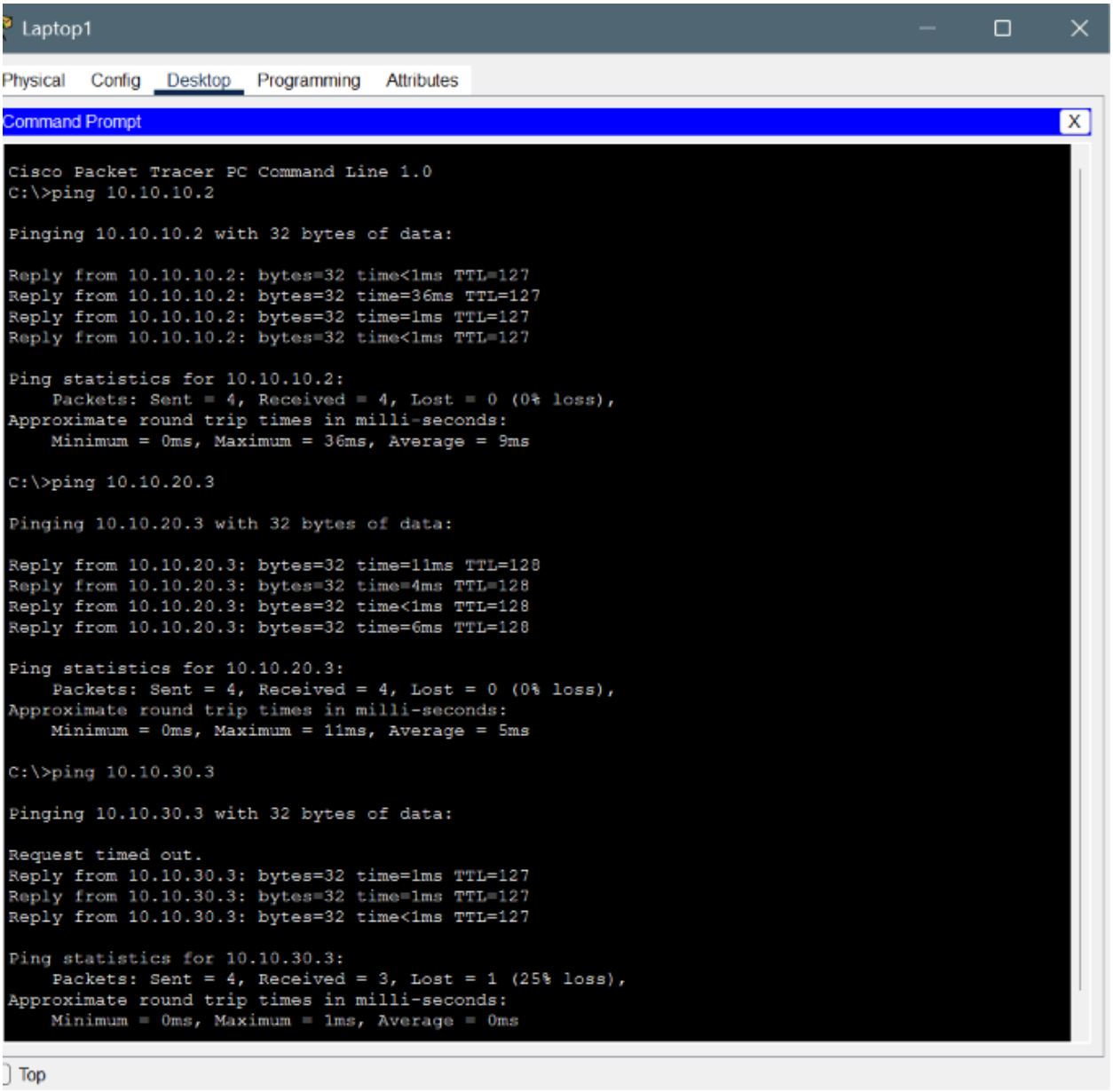
The network was successfully configured and connectivity between the departments was verified using ping tests.

OUTPUT


ip address 10.10.50.0
Gig0/0
Eth1/1
Eth2/1
Eth3/1
Eth7/1
Eth5/1
Fa0
PC-PT
PC0
Laptop-PT
Laptop0
VLAN 10
HR
ip address 10.10.10.0
Gateway 10.10.10.1
PC-PT
PC1
Laptop-PT
Laptop1
VLAN 20
PC-PT
PC2
IT
ip address 10.10.20.0
Gateway 10.10.20.1
PC-PT
PC3
Laptop-PT
Laptop3
VLAN 30
SALE
ip address 10.10.30.0
Gateway 10.10.30.1


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Minimum = 0ms, Maximum = 13ms, Average = 0ms
C:\>ping 10.10.20.2
Pinging 10.10.20.2 with 32 bytes of data:
Request timed out.
Reply from 10.10.20.2: bytes=32 time=15ms TTL=127
Reply from 10.10.20.2: bytes=32 time=21ms TTL=127
Reply from 10.10.20.2: bytes=32 time=22ms TTL=127
Ping statistics for 10.10.20.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 15ms, Maximum = 22ms, Average = 19ms
C:\>ping 10.10.20.3
Pinging 10.10.20.3 with 32 bytes of data:
Request timed out.
Reply from 10.10.20.3: bytes=32 time=1ms TTL=127
Reply from 10.10.20.3: bytes=32 time=1ms TTL=127
Reply from 10.10.20.3: bytes=32 time=1ms TTL=127
Ping statistics for 10.10.20.3:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>ping 10.10.20.3
Pinging 10.10.20.3 with 32 bytes of data:
Reply from 10.10.20.3: bytes=32 time=28ms TTL=127
Reply from 10.10.20.3: bytes=32 time=31ms TTL=127
Reply from 10.10.20.3: bytes=32 time=22ms TTL=127
Reply from 10.10.20.3: bytes=32 time=17ms TTL=127
Ping statistics for 10.10.20.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 17ms, Maximum = 31ms, Average = 24ms
Top


PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
C:\>ping 10.10.30.1
Pinging 10.10.30.1 with 32 bytes of data:
Reply from 10.10.30.1: bytes=32 time<1ms TTL=255
Reply from 10.10.30.1: bytes=32 time=21ms TTL=255
Reply from 10.10.30.1: bytes=32 time=9ms TTL=255
Reply from 10.10.30.1: bytes=32 time<1ms TTL=255
Ping statistics for 10.10.30.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 21ms, Average = 7ms
C:\>ping 10.10.10.1
Pinging 10.10.10.1 with 32 bytes of data:
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time=2ms TTL=255
Reply from 10.10.10.1: bytes=32 time=9ms TTL=255
Reply from 10.10.10.1: bytes=32 time=14ms TTL=255
Ping statistics for 10.10.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 14ms, Average = 6ms
C:\>ping 10.10.10.2
Pinging 10.10.10.2 with 32 bytes of data:
Reply from 10.10.10.2: bytes=32 time<1ms TTL=127
Reply from 10.10.10.2: bytes=32 time=1ms TTL=127
Reply from 10.10.10.2: bytes=32 time=44ms TTL=127
Reply from 10.10.10.2: bytes=32 time=10ms TTL=127
Ping statistics for 10.10.10.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 44ms, Average = 13ms
C:\>
Top


Laptop1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.2
Pinging 10.10.10.2 with 32 bytes of data:
Reply from 10.10.10.2: bytes=32 time<1ms TTL=127
Reply from 10.10.10.2: bytes=32 time=36ms TTL=127
Reply from 10.10.10.2: bytes=32 time=1ms TTL=127
Reply from 10.10.10.2: bytes=32 time<1ms TTL=127
Ping statistics for 10.10.10.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 36ms, Average = 9ms
C:\>ping 10.10.20.3
Pinging 10.10.20.3 with 32 bytes of data:
Reply from 10.10.20.3: bytes=32 time=11ms TTL=128
Reply from 10.10.20.3: bytes=32 time=4ms TTL=128
Reply from 10.10.20.3: bytes=32 time<1ms TTL=128
Reply from 10.10.20.3: bytes=32 time=6ms TTL=128
Ping statistics for 10.10.20.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 5ms
C:\>ping 10.10.30.3
Pinging 10.10.30.3 with 32 bytes of data:
Request timed out.
Reply from 10.10.30.3: bytes=32 time=1ms TTL=127
Reply from 10.10.30.3: bytes=32 time=1ms TTL=127
Reply from 10.10.30.3: bytes=32 time<1ms TTL=127
Ping statistics for 10.10.30.3:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
Top